ASTRA piNas

CM5 LITE / DUAL 2280 NVMe / GIGABIT ETHERNET

REVIEW DRAFT — POE SOURCING AND MECHANICAL VERIFICATION OPEN
Unrouted placement study. Do not fabricate. See docs/REVIEW.md.

01 / CM5 Lite interface

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02 / PCIe switch and clock distribution

File: 02_pcie.kicad_sch

03 / PCIe core power and bypassing

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08 / microSD and USB-UART

File: 08_sd_uart.kicad_sch

Project-local libraries / LCSC sourcing	
Sheet: /	
File: astra_piNas.kicad_sch	
Title: Astra piNas / CM5 dual NVMe carrier	
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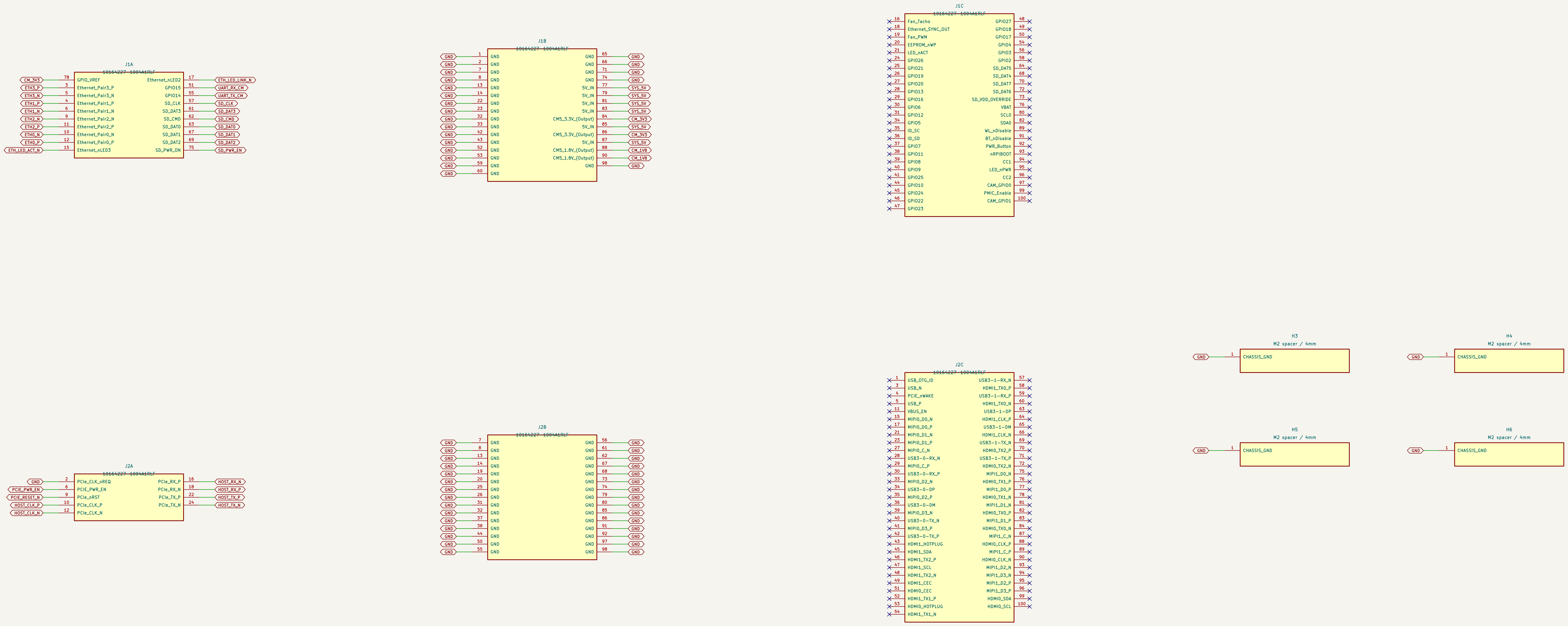
01 / CM5 Lite interface

4 mm official Amphenol connectors. CM5 Lite required for native microSD. No eMMC module.
Connector J2 local pin 1 corresponds to module pin 101. UART uses GPIO14/15 with OS configuration.
CM5 PCIe receive P/N intentionally reversed for routing; automatic receiver polarity correction is supported.



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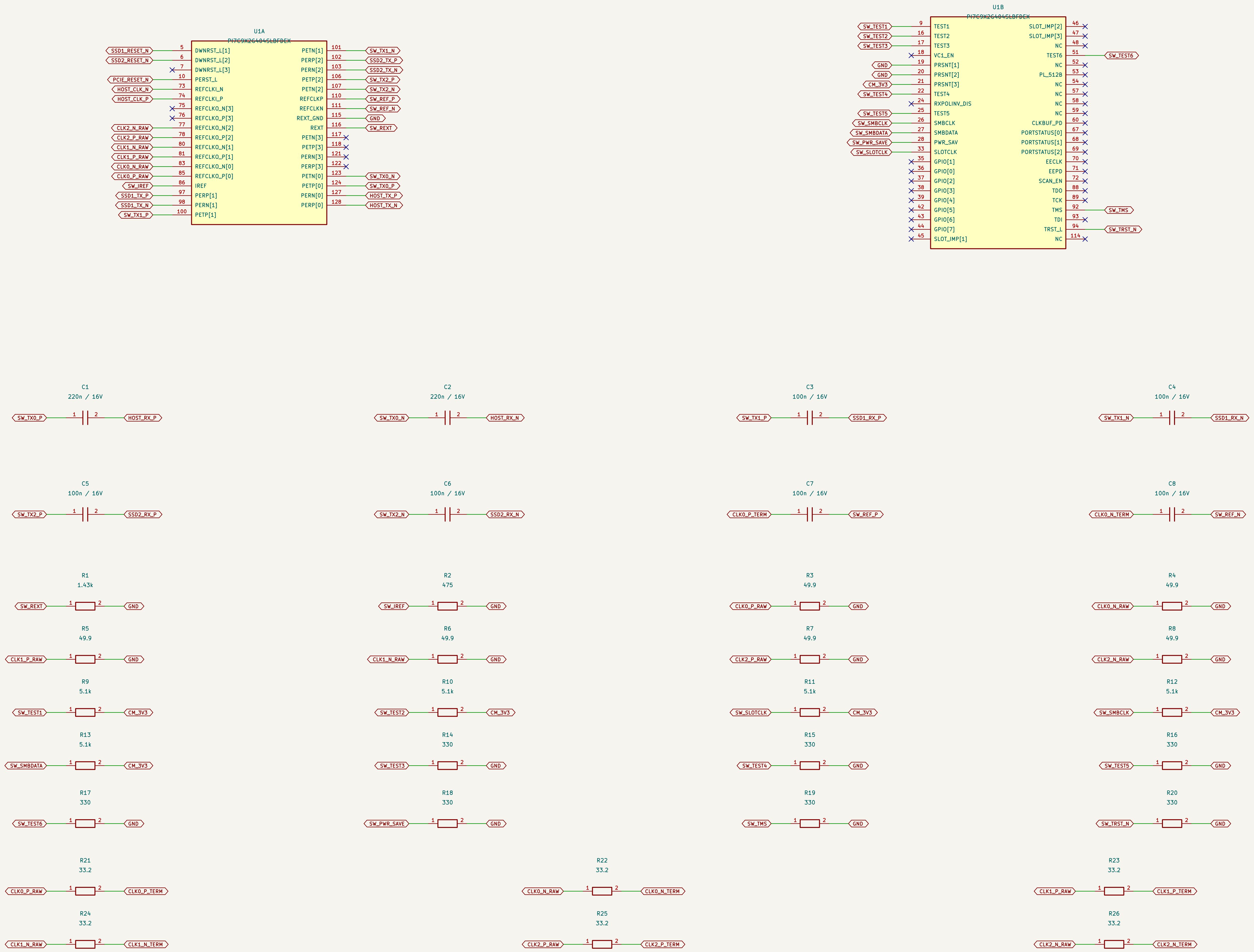
02 / PCIe switch and clock distribution

One upstream x1 Gen 2 link shared by two x1 NVMe links. Port 3 absent. Integrated HCSL clock buffer enabled.
Defaults intentionally left open where the datasheet permits internal straps. No EEPROM required for default enumeration.
Upstream RX P/N intentionally reversed. RXPOLINV_DIS remains unconnected, using its internal pull-down.



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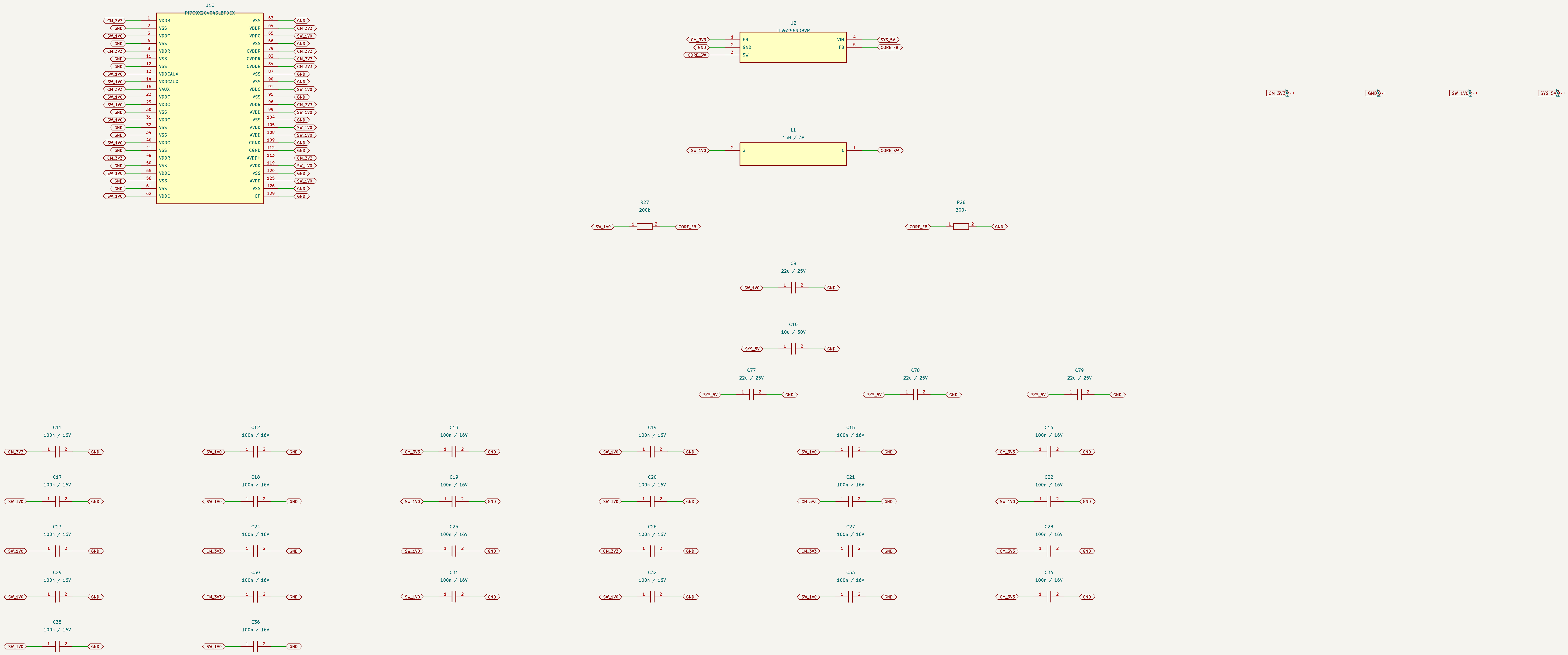
03 / PCIe core power and bypassing

1.0 V regulator enabled by CM5 3.3 V: I/O supply precedes switch core. Keep each bypass capacitor at its associated supply pin.
CM5 3.3 V auxiliary load includes switch I/O, clock buffer, microSD and UART I/O; never use it to power SSDs.



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File: 03_aux_power.kicad_sch

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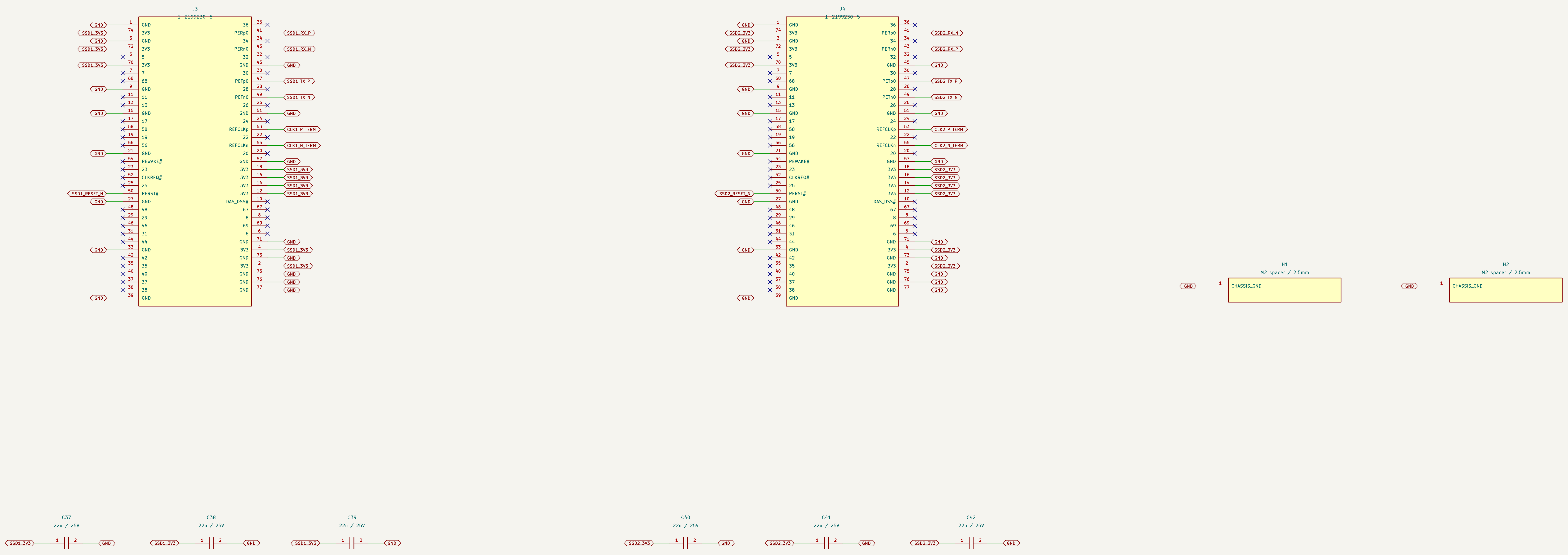
04 / Two M.2 2280 NVMe sockets

NVMe / PCIe SSDs only. SATA M.2 devices are not supported. Both drives run at x1 and share the host Gen 2 x1 link.
Three 22u capacitors at each socket supplement regulator output capacitors. 2280 retention hardware is a separate mechanical item.
SSD2 receive P/N intentionally reversed at the M.2 connector; PCIe receiver polarity correction is supported.



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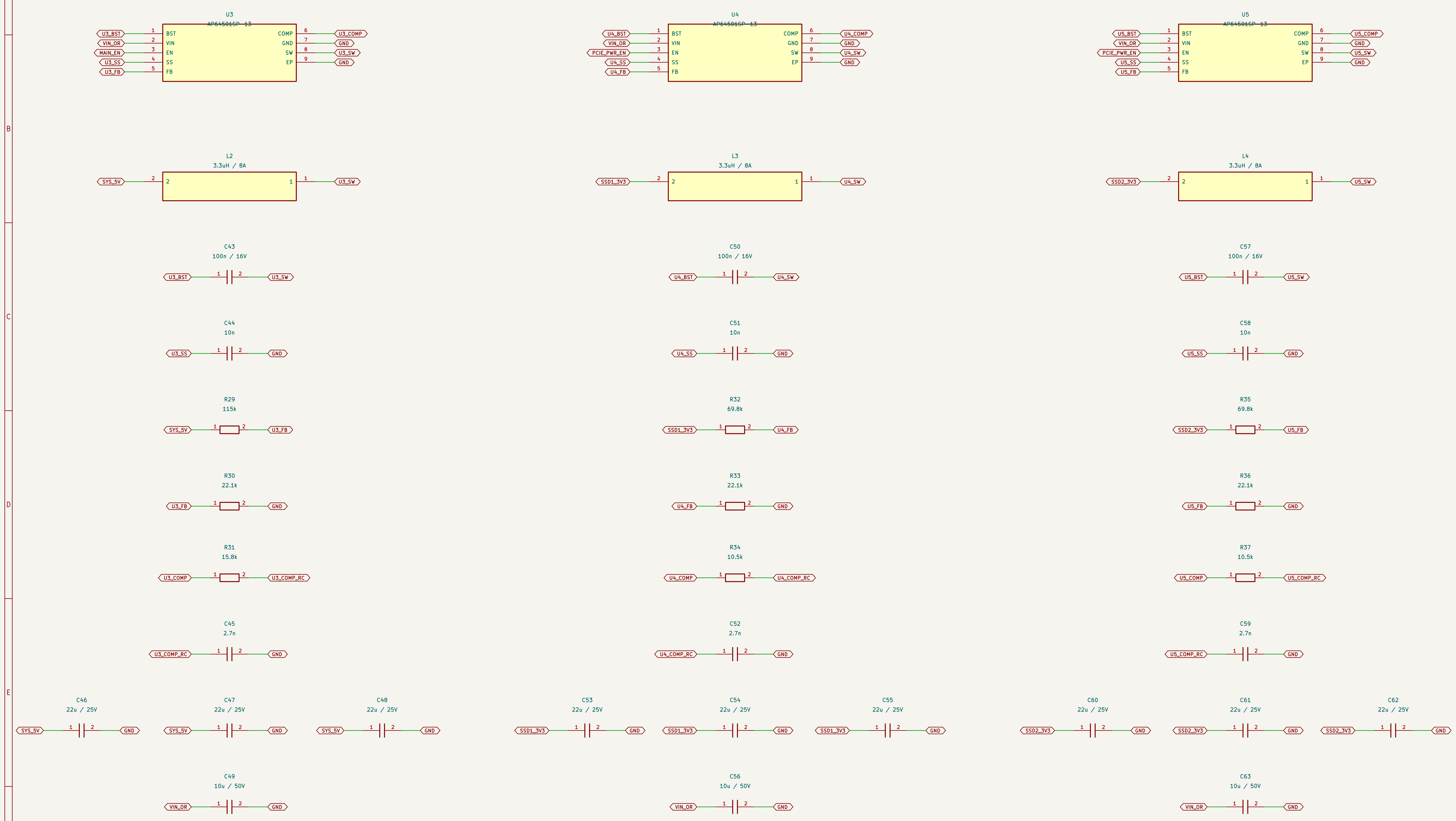
05 / Main 5 V and independent SSD power rails

AP64501, 40 V rated synchronous bucks. 5 V input regulator UVLO ~6.5 V; SSD converters enabled by CM5 PCIe power enable.
SSD supply design target: 3 A per socket. Overall available power remains limited by the selected input source and thermal design.



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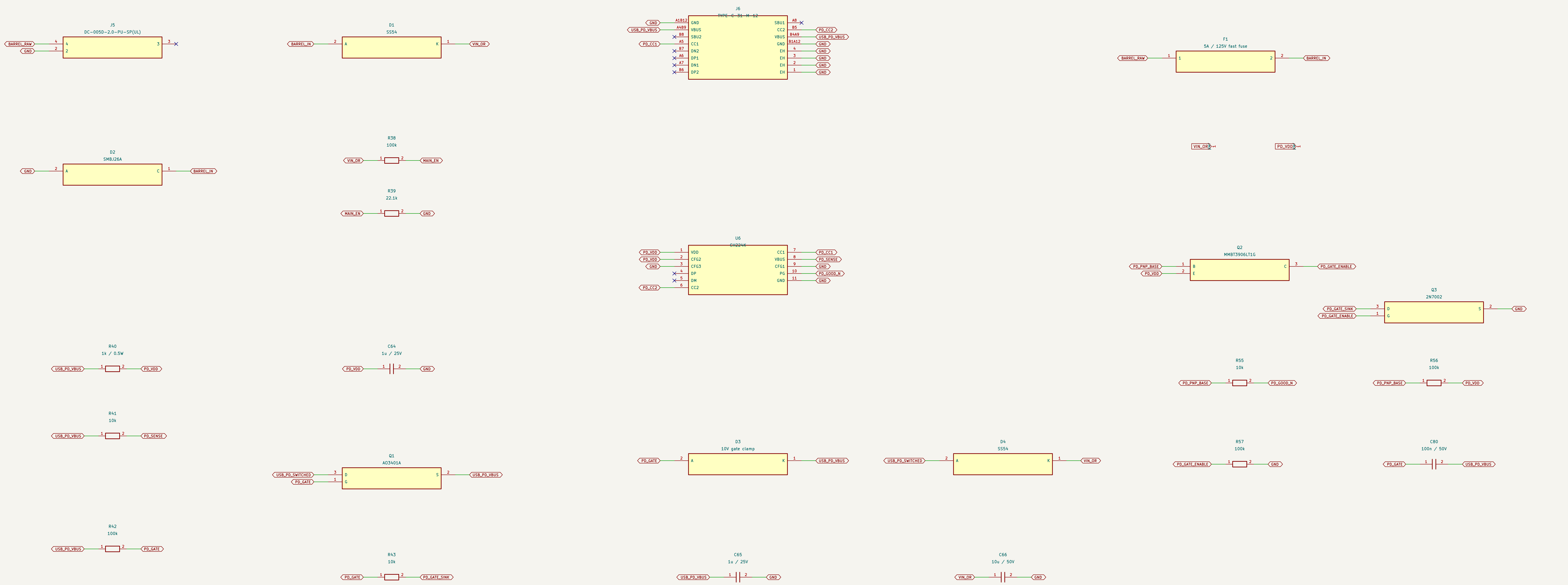
06 / Barrel and USB–C PD power inputs

Center–positive barrel input: 9–25 V DC. USB–C requires a 20 V / 3 A PD supply; its data contacts are unconnected.
Schottky source OR–ing prevents reverse feed. The highest source normally supplies the load; seamless changeover is not guaranteed.



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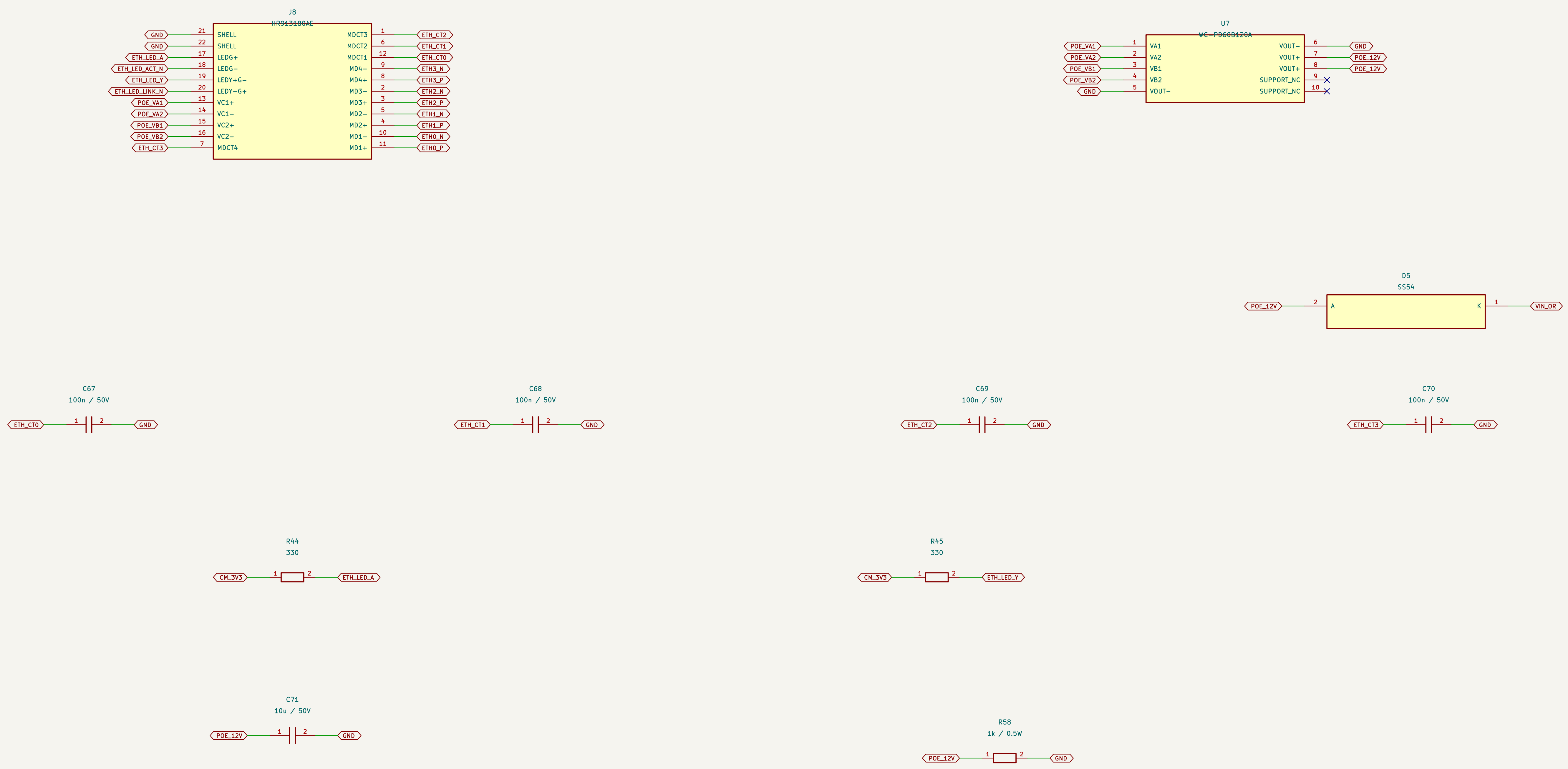
07 / Gigabit Ethernet and PoE++ candidate

PoE module is a PROVISIONAL sourcing choice: only 15 in LCSC stock at capture. Low stock does not satisfy the original deep-stock requirement.
Do not release this sheet for manufacture until module pin 9/10 mechanical coordinates and 802.3bt class power are confirmed.



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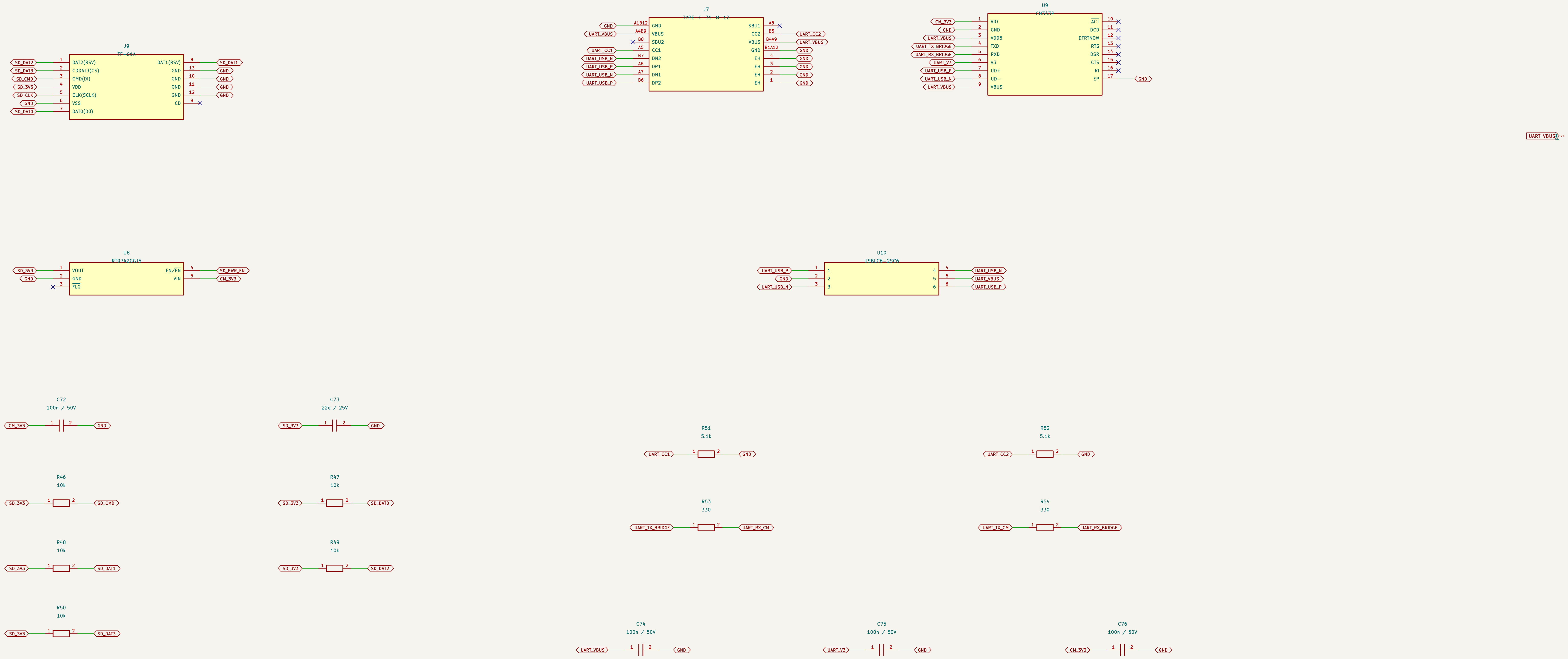
08 / microSD and USB–UART

USB–UART is independently powered by its USB cable; VIO follows CM5 3.3 V. No connection between debug VBUS and SYS_5V.
GPIO14/15 UART requires OS configuration; this is not the separate CM5 on–module boot–debug UART.



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